

Comparison of Random Graph Models

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Introduction

We investigate three random graph models: Erdős–Rényi model, Watts–strogatz model and Kleinberg model.

- 1 Erdős–Rényi model $G(n, p)$: $V(G) = [n]$. Every pair is included in $E(G)$ with probability p , independently.
- 2 Watts–strogatz model: Given $N, K \in \mathbb{N}$, $p \in (0, 1)$. First construct a regular cycle with N nodes, and each connected to K neighbors, $K/2$ on each side. For every node $i = 0, 1, \dots, n-1$, we take every edge connecting i to its $K/2$ rightmost neighbors and rewire it with probability p . Rewiring means replacing (i, j) by (i, k) where k is chosen uniformly at random from all possible nodes while avoiding $k \neq i$ and multiple edges.

Introduction

- ① Kleinberg model: Given $N \in \mathbb{N}^*$, $r \geq 0$. Let $V(G) = (\mathbb{Z}/N\mathbb{Z})^d$. We first connect an undirected edge to its neighbors of d -dim lattice. Then, for any non-adjacent pair $\{u, v\}$, we connect them with probability $\alpha_u \cdot d(u, v)^{-r}$, where $d(u, v)$ represents the distance between u and v on the grid, and we define

$$\alpha_u := \left(\sum_{v \text{ non adjacent to } u} d(u, v)^{-r} \right)^{-1}$$

for some fixed $u \in V$.

For each model, we will investigate the properties of degree distribution, size of all components, diameter and average distance of a giant cluster. For the last two models, we also study their clustering coefficient, and performance of navigating algorithms on the model.

Erdős–Rényi model $G(n, \lambda/n)$

- Consider $p = \lambda/n$ with constant $\lambda > 0$.
- Phase transition at $\lambda = 1$:
 - $\lambda < 1$: subcritical (many small components)
 - $\lambda = 1$: critical (largest component $\sim n^{2/3}$)
 - $\lambda > 1$: supercritical (giant component appears)

Degree distribution

Theorem (Bollobás, 1981)

For $G(n, \lambda/n)$ as $n \rightarrow \infty$, the empirical degree distribution converges to $\text{Poisson}(\lambda)$:

$$\frac{1}{n} \#\{i : \deg(i) = k\} \xrightarrow{P} e^{-\lambda} \frac{\lambda^k}{k!}.$$

Numerical experiment ($n = 10^5$, $\lambda = 0.5, 1, 2$):

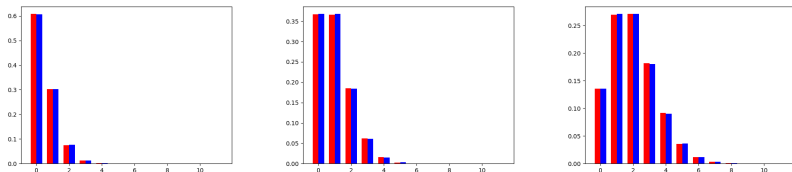


Figure: Poisson approximation (blue) vs. observed (red)

Theorem (Component sizes)

Let L_1 be the largest component size of $G(n, \lambda/n)$.

- If $\lambda < 1$, then we have $L_1 / \log n \rightarrow 1/I(\lambda)$, where $I(\lambda) = \lambda - 1 - \log \lambda$.
- If $\lambda = 1 + cn^{-1/3}$, then we have $n^{-2/3} L_1 \xrightarrow{d} \mathcal{L}_c$, where $\mathcal{L}_c$ is a non-trivial random variable.
- If $\lambda > 1$, then we have $\frac{L_1}{n} \xrightarrow{d} 1 - \rho$, where $\rho = e^{\lambda(\rho-1)}$.
Moreover, all other components are of the size $O(\log n)$.

Numerical simulation of component sizes

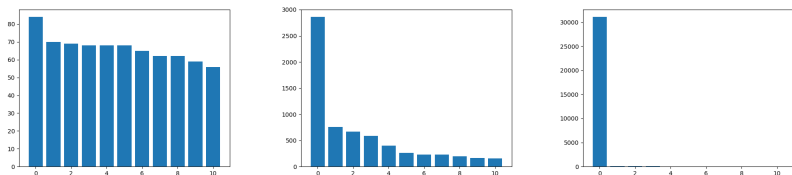


Figure: Largest $\lfloor \log n \rfloor$ components; $\lambda = 0.8, 1.0, 1.2$

Diameter and average distance

Theorem (Diameter and average distance)

Let D_λ and $\bar{L}_\lambda$ be the diameter and average distance of the largest cluster of $G(n, \lambda/n)$.

- If $\lambda < 1$, then we have $\frac{D_\lambda}{\log n / \log(1/\lambda)} \xrightarrow{P} 1$ and $\frac{\bar{L}_\lambda}{\log n / \log(1/\lambda)} \xrightarrow{P} 1$.
- If $\lambda = 1 + cn^{-1/3}$, then $n^{-1/3} D_\lambda$ and $n^{-1/3} \bar{L}_\lambda$ converge to a non-trivial distribution depending on c .
- If $\lambda > 1$, then we have $\frac{D_\lambda}{\log n / \log \lambda} \xrightarrow{P} 1$ and $\frac{\bar{L}_\lambda}{\log n / \log \lambda} \xrightarrow{P} 1$.

Simulation results for diameter and average distance

(n, λ)	Diameter D	Average $\bar{L}$
$(100000, 0.8)$	25	20.16
$(100000, 1)$	120	93.45
$(2000, 1.2)$	53	40.2

Table: Experimental values

Results match the theoretical order: $\log n / \log(1/\lambda)$ for subcritical, $n^{1/3}$ for critical, $\log n / \log \lambda$ for supercritical.

Watts–Strogatz Model: Introduction

The Erdős–Rényi graph does not have two important properties observed in many real-world networks: (1) it does not generate local clustering (has low clustering coefficient); (2) it does not account for the formation of hubs (degree distribution does not follow a power law, which is observed in many real-world, scale-free networks). The Watts–Strogatz model was designed to address the **first** limitation.

- $p = 0$: regular lattice (high clustering, large diameter).
- $p = 1$: random graph-like (low clustering, short diameter).
- $0 < p \ll 1$: **small-world regime** – both high clustering and short paths.

Watts-Strogatz Model: Clustering Coefficient

Definition (Local clustering coefficient)

For a vertex v with degree $\deg(v) \geq 2$, let e_v be the number of edges among its neighbors. Then the local clustering coefficient C_v is defined as

$$C_v := \frac{2e_v}{\deg(v)(\deg(v) - 1)}.$$

If $\deg(v) \leq 1$, $C_v = 0$.

Definition (Global clustering coefficient)

The global clustering coefficient C is defined as

$$C := \frac{\#\{\text{triangles in the graph}\}}{\#\{\{v_1, v_2, v_3\} \subseteq V : \text{at least two of three pairs are connected}\}}$$

Clustering Coefficient and diameter

Theorem (Watts & Strogatz 1998)

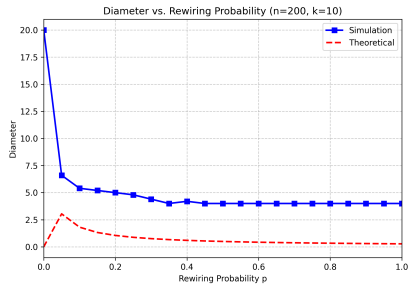
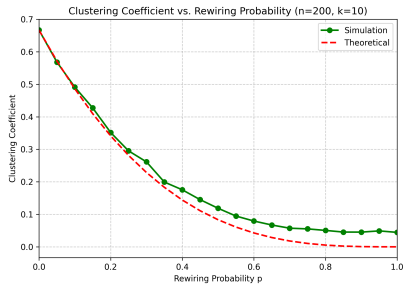
For the initial lattice ($p = 0$), every vertex has the same local clustering coefficient: $C(0) = \frac{3(k-2)}{4(k-1)}$.

For small p , a widely used approximation is $C(p) \approx C(0) \cdot (1 - p)^3$.

Remark

There is no specific results about the diameter. However, there exist an approximation result about the average path length (using mean field approximation). Let $\ell(n, p, k)$ be the average shortest-path length, then $\ell(n, p, k) \approx \frac{n}{2k} F(pkn)$, where $F(\cdot)$ is a function satisfies $F(x) \sim \frac{\log x}{x}$ as $x \rightarrow \infty$.

Simulation of Watts-Strogatz model



Kleinberg Model: introduction

The famous "six degrees" theory states that, all people are six or fewer social connections away from each other. This actually implies something stronger than a single diameter property of the network. In Milgram's small-world experiment, people were asked to deliver letters to others without knowing their detailed personal information. In this experiment, nobody in that social experiment had a "God's eye view", every recipient had to make the decision based on their own information!

In this case, Watts-Strogatz model fails to account for this phenomenon. This is the reason why Kleinberg model was proposed.

Kleinberg Model: Diameter

Theorem (Diameter phase transition)

For the d -dimensional Kleinberg grid model, the expected diameter exhibits a phase transition in the exponent r :

$$\mathbb{E}[\text{diam}(G_n)] = O((\log n)^{\beta(d,r)}) \quad \text{for } r < 2d,$$

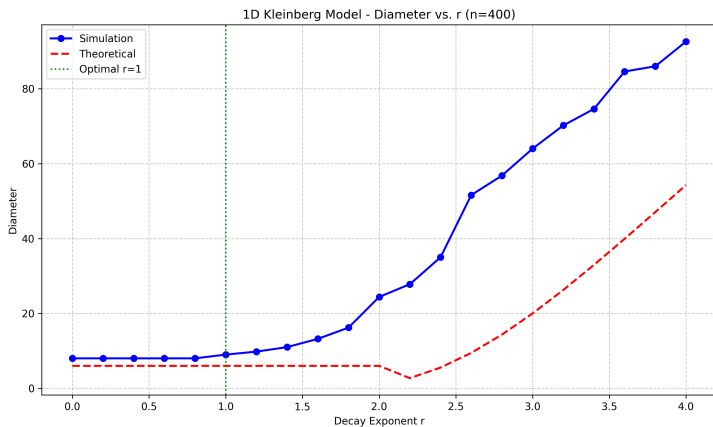
for some constant $\beta(d, r) > 0$, while for $r > 2d$ one has the polynomial lower bound

$$\mathbb{E}[\text{diam}(G_n)] = \Omega\left(n^{\frac{r-2d}{r-d}-o(1)}\right).$$

Especially, for $r \leq d$,

$$\mathbb{E}[\text{diam}(G_n)] = \Theta(\log n).$$

Kleinberg Model: Simulation of Diameter



Decentralized Routing

Definition (Decentralized algorithm)

At each step, the current message holder may use only:

- the target's coordinates,
- the current vertex's coordinates and its incident edges,
- any local conventions (e.g., random number generator).

It cannot access the global network structure.

Definition (Greedy algorithm)

Starting from source s , at each step move to the neighbor whose lattice distance to the target t is minimized.

Greedy Algorithm Details

Definition (Greedy algorithm for Kleinberg model)

- 1 Let current vertex $u = s$.
- 2 While $u \neq t$:
 - Among all neighbors of u ($2d$ grid neighbors + one long-range contact), pick v that minimizes $d(v, t)$.
 - Go to the adjacent vertex v , let $u \leftarrow v$.

This uses only local information: the current vertex's neighbor list, the target's coordinates, and Manhattan distance.

Upper Bound: Greedy Routing in 2D

Theorem (Kleinberg 2000)

In the two-dimensional grid ($d = 2$) with $r = 2$ and one long-range contact per vertex, the greedy algorithm finds a path from s to t in expected time

$$\mathbb{E}[T] = O((\log n)^2).$$

- The exponent $r = 2$ matches the dimension $d = 2$.
- The bound is polylogarithmic – efficient even for large n .

Lower Bound for Any Decentralized Algorithm

Theorem (Kleinberg 2000)

*For the d -dimensional Kleinberg model, the expected delivery time of **any** decentralized algorithm satisfies*

$$\mathbb{E}[T] \geq \alpha_{d,r} n^{c(d,r)},$$

where $\alpha_{d,r} > 0$ is a constant and

$$c(d, r) = \begin{cases} \frac{d-r}{d+1}, & 0 \leq r < d, \\ \frac{r-d}{r-1}, & r > d. \end{cases}$$

Comparison of the three models

Model	Diameter	Clustering	Navigable
Erdős–Rényi (supercritical)	$O(\log n)$	Very low	No
Watts–Strogatz	$O(\log n)$	High	No
Kleinberg ($r = d$)	$O(\log n)$	High	Yes

Table: Comparison of three models

- Erdős–Rényi: lacks clustering and metric structure.
- Watts–Strogatz: high clustering, small diameter, but not navigable.
- Kleinberg: preserves clustering and small diameter, and *is* navigable when $r = d$.

Conclusion

- Erdős–Rényi model: simple but unrealistic (no clustering, no geometry).
- Watts–Strogatz model: adds clustering, keeps short paths, but still not navigable.
- Kleinberg model: introduces a distance-dependent long-range link distribution; achieves **navigability** when $r = d$.

Kleinberg's result shows that the *right* amount of spatial bias (r equal to the lattice dimension d) makes decentralized search efficient.

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